M/J Comprehensive Science 1

Grade 6 · Course #2002040 · Full Year (Semesters 1 & 2) · Optima Academy Online

Family Course Overview
A one-page guide to the year ahead

Comprehensive Science 1 is a yearlong survey of **life, physical, and Earth & space science** that trains sixth graders to observe nature with wonder and precision, then reason carefully from evidence. Students keep a science notebook all year, build and create constantly — museums, machines, mission reenactments — and finish the year as young engineers planning humanity's return to the Moon.

THE YEAR AT A GLANCE

Q1 Life Science — ~8 weeks

Cells and the characteristics of living things, classification of organisms, and a tour of the human body's systems — ending with germs, the immune system, and a quarter reflection.

> **Project:** Operation Taxonomy — students become classifiers and sort the living world the way scientists do.

Q2 Energy & Forces — ~9 weeks

The Law of Conservation of Energy, potential vs. kinetic energy, then forces: free-body diagrams, balanced and unbalanced forces, and gravitation, closing with a quarter test.

> **Project:** Rube Goldberg — "The Endless Energy Machine," an over-engineered contraption that shows energy changing forms.

Q3 Earth & Space Science — ~9 weeks

How the Earth maintains life: the spheres of the Earth, heat and energy, weather and hurricanes — connecting global patterns back to Florida.

Q4 Space Exploration & Engineering — ~9 weeks

The solar system and beyond: NASA's Artemis program, moon-base design, and the engineering mindset, with end-of-year quizzes wrapping up the journey.

> **Projects:** Planet Travel Project — research a planet and record a "travel sales pitch" — plus an Artemis II Mission Reenactment.

A TYPICAL WEEK

- **Read:** a weekly overview page and Quick Reference Guide(s) introducing the concept
- **Do:** 1–3 weekly assignments — science notebook entries, drawings, models, and creative builds
- **Explore:** VR experiences on Optima's virtual campus (photos & recordings submitted in Canvas)
- **Check in:** quizzes and tests at the end of each unit and quarter

TIME COMMITMENT

250 minutes / week ≈ 4 hrs 10 min

- Lessons & reference guides: ~60 min
- Weekly assignments & notebook: ~120 min
- VR activities & exploration: ~45 min
- Quizzes, discussions, review: ~25 min

Weekly minutes come from Optima's master schedule; the per-component split is an estimate. Project weeks (Rube Goldberg, Planet Travel) may run longer.

WHAT STUDENTS NEED

- VR headset for Optima's virtual campus
- Science notebook, pencils, highlighters
- Art supplies for projects (lists provided well before each due date)
- **On AI:** Optima embraces AI as a tool that enhances — never replaces — learning, guided by the seven Pillars of Virtue

ON-DEMAND (ASYNCHRONOUS)

This course is designed for self-paced learning: students work through the weekly modules on their own schedule with a teacher available for support, feedback, and grading.

LIVE SCHOOL

Live-school students complete the same course in VR every school day, together with a teacher and classmates in real time.

PLEASE NOTE

This overview describes the course as designed. The week-to-week experience may vary from teacher to teacher, as each teacher leans into their own strengths and passions.